Competitive Exam Practice 1 — Answer Key

Computer Architecture, Organization, and Performance

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Q1 **1.6.** Same ISA and same program ⇒ same instruction count I. time $= I \cdot \mathrm{CPI}/f$. P_2's time is $0.75 \times P_1$'s and its CPI is $1.2\times$:

$$\frac{t_2}{t_1} = \frac{\mathrm{CPI}_2/f_2}{\mathrm{CPI}_1/f_1} = \frac{1.2}{f_2/1} = 0.75 \quad \Rightarrow \quad f_2 = \frac{1.2}{0.75} = 1.6 \text{ GHz}.$$

Q2 **3.** Serial part (10% of 100 ns) = 10 ns. Parallel part = 90 ns, split over n cores. Overhead for $n-1$ additional cores $= 10(n-1)$ ns. Total:

$$T(n) = 10 + \frac{90}{n} + 10(n-1) = \frac{90}{n} + 10n.$$

Minimize: $\frac{dT}{dn} = -\frac{90}{n^2} + 10 = 0 \Rightarrow n^2 = 9 \Rightarrow n = 3$ (integer minimizer). $T(3) = 30 + 30 = 60$ ns.

Q3 **4.** Non-pipelined: each instruction takes 6 cycles (all stages serial). For N instructions: $6N$ cycles. Pipelined: ideal CPI = 1; 25% of instructions add 2 stalls each ⇒ average CPI $= 1 + 0.25 \times 2 = 1.5$. Pipelined cycles $= 1.5N$. Speedup $= \frac{6N}{1.5N} = 4$.

Q4 **2.50.** Non-pipelined time for N instructions: $N \times 5$ cycles at 1.6 GHz $= \frac{5N}{1.6}$ ns. Pipelined: CPI $= 1 + 0.30 \times 2 = 1.6$; clock 1.2 GHz ⇒ time $= \frac{1.6N}{1.2}$ ns. Speedup $= \frac{5N/1.6}{1.6N/1.2} = \frac{5\times1.2}{1.6\times1.6} = \frac{6}{2.56} \approx 2.34$. Recompute carefully: $\frac{5}{1.6} \cdot \frac{1.2}{1.6} = \frac{6}{2.56} = 2.34375 \approx \mathbf{2.34}$. (Note: the 1G $= 10^9$ prefix cancels between numerator and denominator, so only the cycle ratios and clock ratios matter.)

Q5 **(C).** For the *same* clock, execution time $\propto$ CPI, not MIPS standing alone. Both machines run at 1 GHz; M_1's CPI 1.25 gives time $\propto 1.25$, M_2's CPI 2.5 gives time $\propto 2.5$ — but with identical f, the actual times are equal when the instruction counts differ correspondingly. The core point is (C): time is f/CPI-determined and f is equal, so the reported MIPS figures (which embed different instruction counts) are not evidence of a speed difference. (D) states the caveat for differing ISAs, but the question pins both to the same program at the same clock, so (C) is the correct single answer: the machines run the program in equal time.

Q6 **(B).** Amdahl's Law: speedup $= 1/[(1-f) + f/S_f]$. As $S_f \to \infty$, speedup $\to 1/(1-f) = 1/0.70 \approx 1.43\times$. The 70% unimproved fraction sets an unbreakable ceiling, so (A) is false and (C)/(D) misstate the ceiling.